

CS & IT ENGINEERING



Computer Network

TCP Connection Establishment

Lecture No. - 01



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Topics to be Covered



Topic

TCP Connection Establishment



CS-24

CS-25

CS-26

CS-27 $\Rightarrow$?

ABOUT ME

Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

My Telegram Channel

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Topic : Syllabus



V.V. Imp. (AIMD)
[CS-26, sel-1, sel-2]

→ TCP - flow control and congestion control, socket API;

Socket

* Socket
(Network)
Programming

4PYQ

Flow Control Protocols

(1) stop-and-wait (CS-25)

(2) Sliding Window (CS-21)

(a) Go Back N-ARQ

(b) selective Repeat ARQ

* UDP

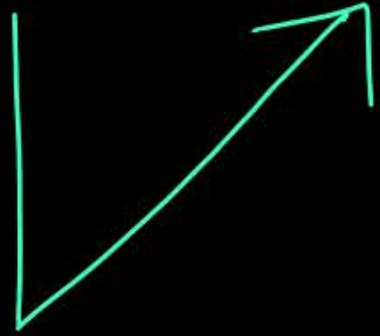


Topic : Transport Layer



Two Transport Layer Protocols :

1. UDP : User Datagram Protocol
2. TCP : Transmission Control Protocol





Topic : UDP



→ Provide 'Connection-less' and 'Unreliable' services to application processes
(Network Processes)

*UDP : Basic Protocol

↳ Simple and Fast



Topic : TCP



→ Provide 'Connection-oriented' and 'Reliable' services

* $TCP = UDP + \text{Extra Services}$

↳ Complex and slow



Topic : Two Process Communication

Source Host

Application Layer

P₁

→ client
web browser
(web client)

(P₁ & P₂)

Two network
process

Destination Host

Application Layer

P₂

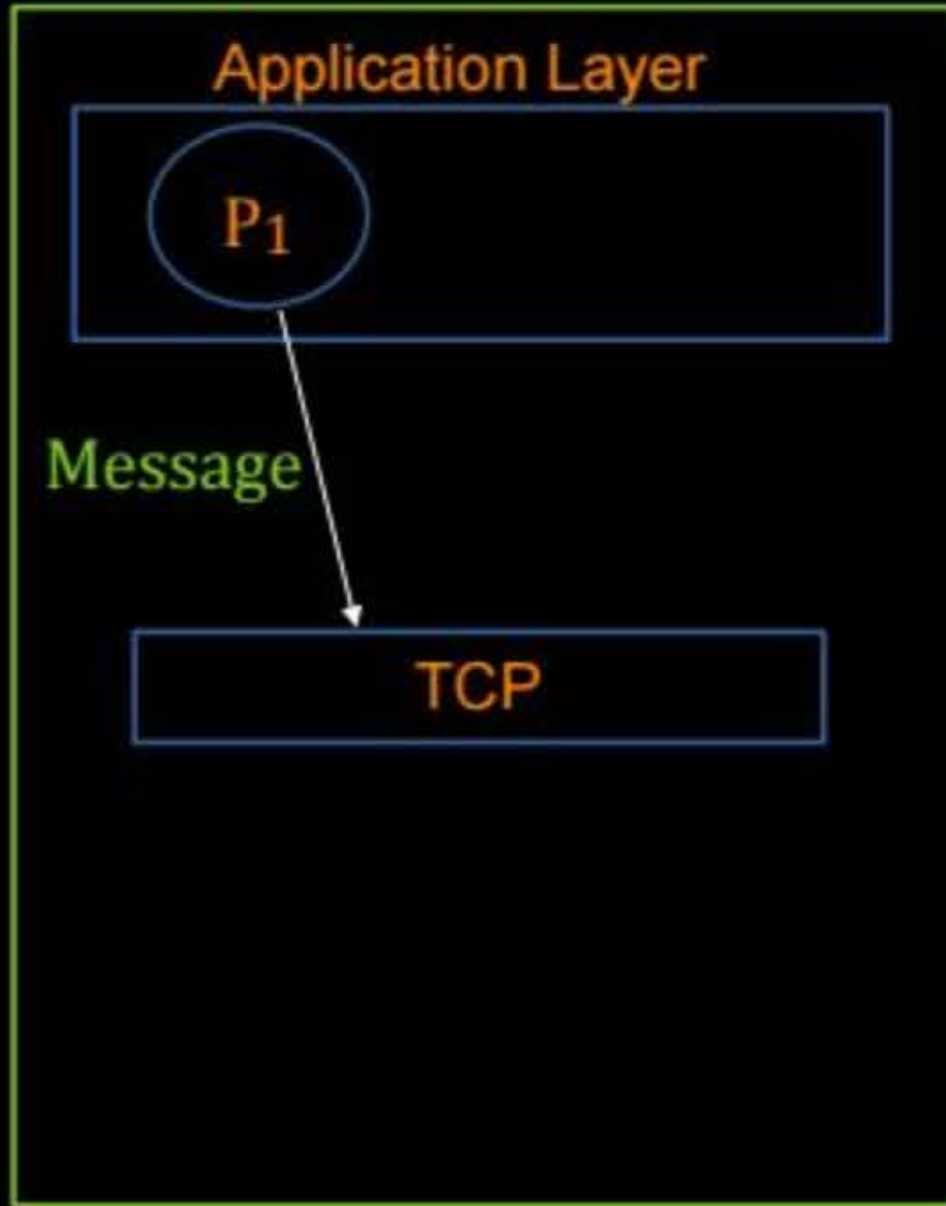
server
(google) ←
* web server



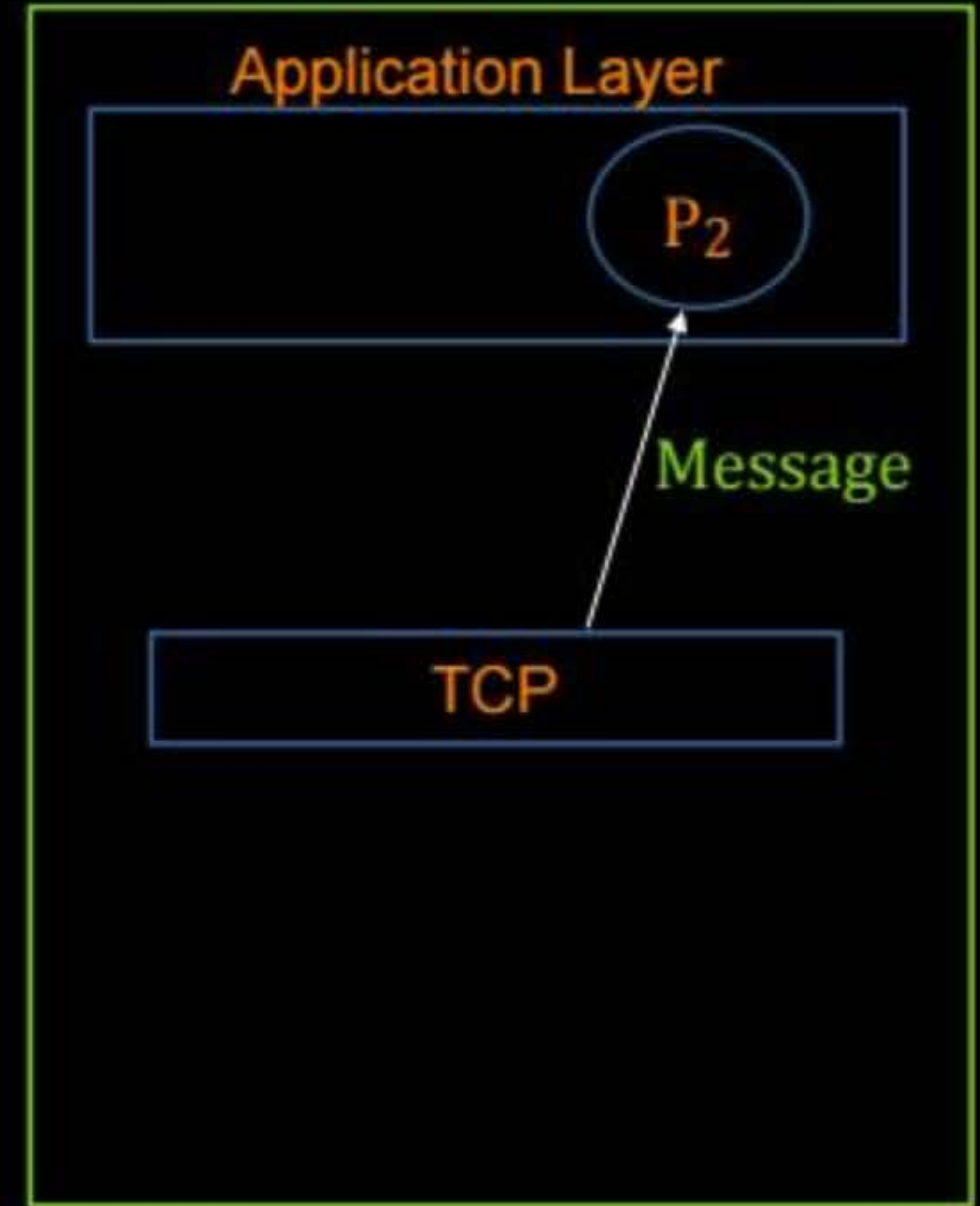
Topic : TCP



Source Host



Destination Host

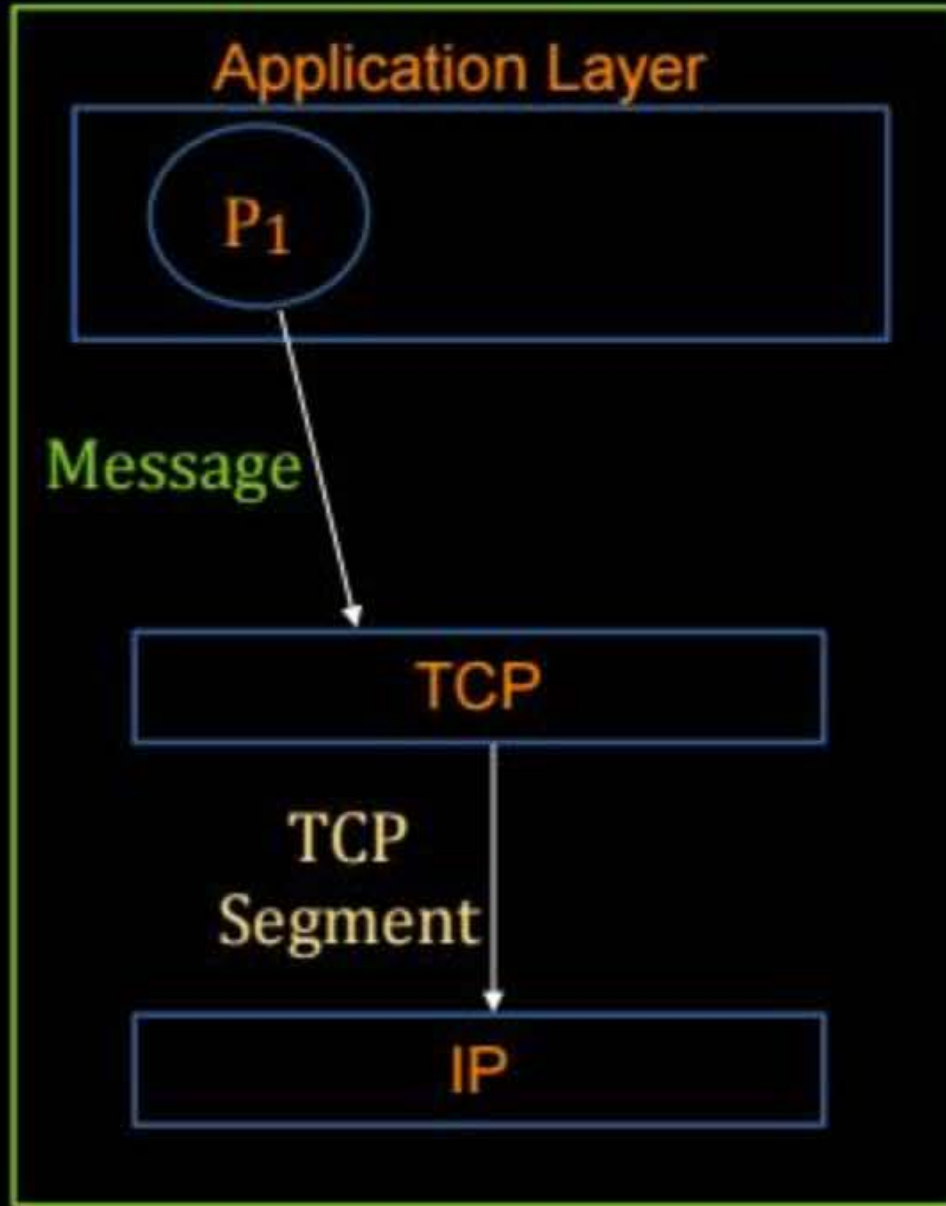




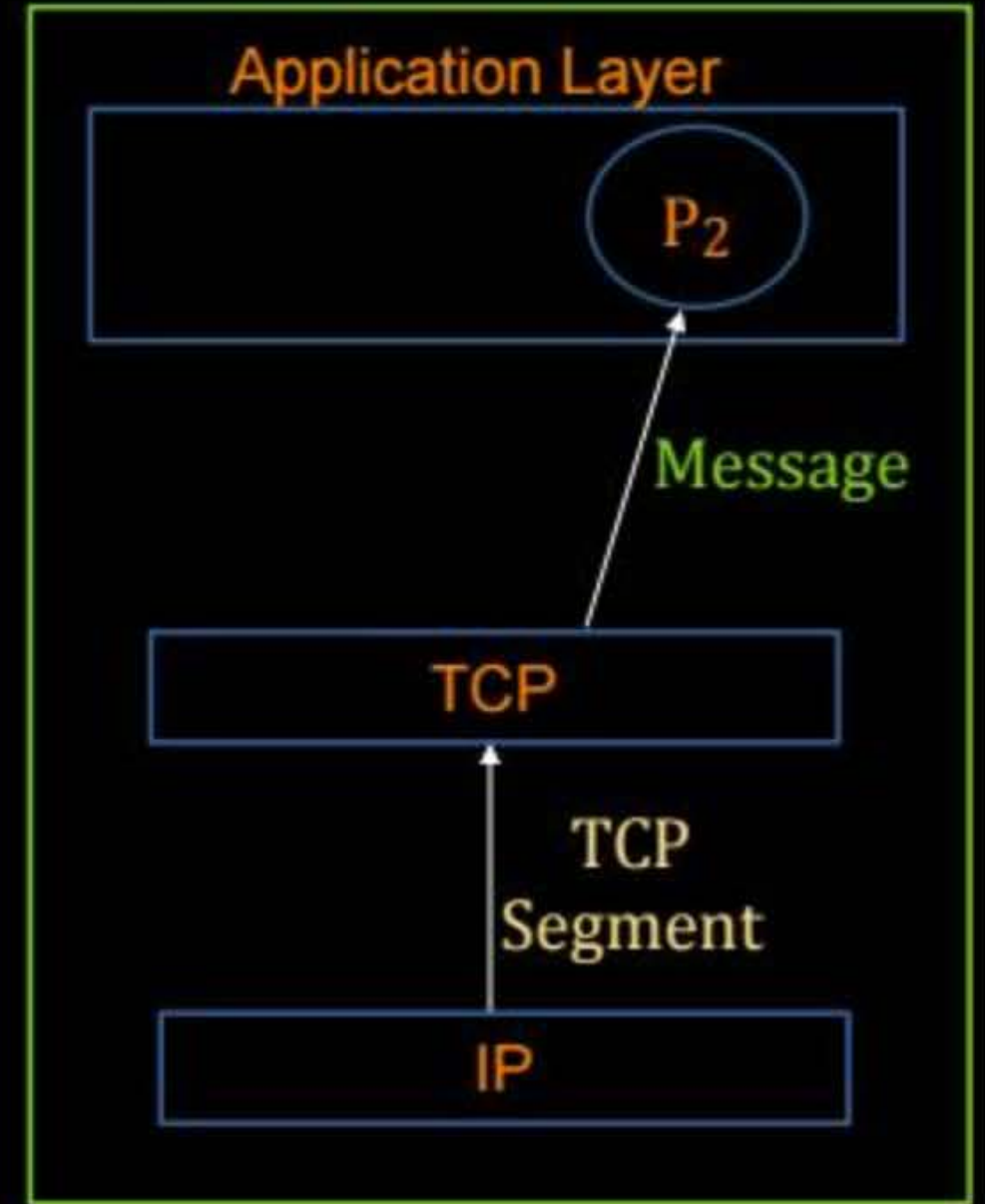
Topic : TCP



Source Host



Destination Host





Topic : TCP

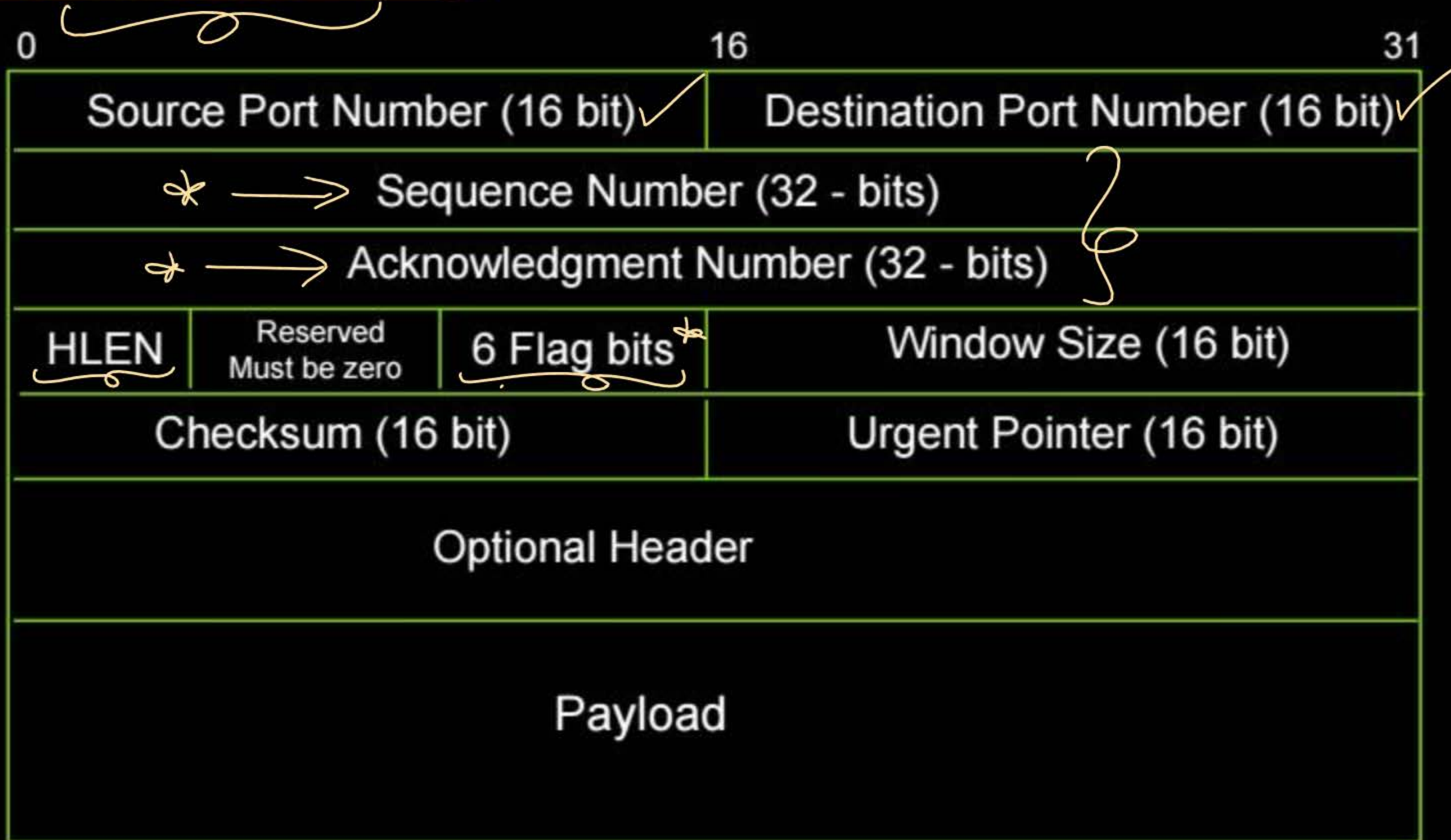


=> Connection-oriented :

-> Need to establish (logical) connection between TCP sender and receiver



Topic : TCP Header





Topic : TCP Header

$P_1(I \subset P) \rightarrow \text{TCP Segment}$



$\Rightarrow$ Source port number : 16-bits $\rightarrow P_1 \text{ Port No.}$

$\Rightarrow$ Destination port number : 16-bits $\rightarrow P_2 \text{ Port No.}$



Topic : TCP Header



=> Sequence Number : 32-bits

=> Acknowledgment Number : 32-bits



Topic : TCP Flag Bits



→ ON/OFF
1/0
set/Reset

=> URG : Urgent

=> ACK : Acknowledgement ***

=> PSH : Push

=> RST : Reset *

=> SYN : Synchronize → TCP Conn. Estb

=> FIN : Finish → TCP Conn. close



Topic : TCP Operation



Three phases of TCP operation :

1. Connection establishment
2. Data transfer
3. Connection termination



Topic : TCP Connection Establishment

→ Connection establishment between TCP client and TCP server





Topic : TCP Connection Establishment

→ Always TCP client initiate the connection request to TCP server

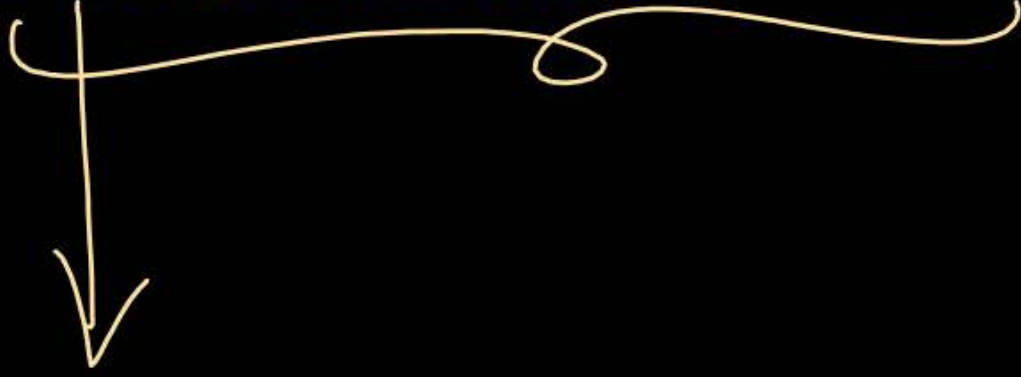




Topic : TCP Connection Establishment

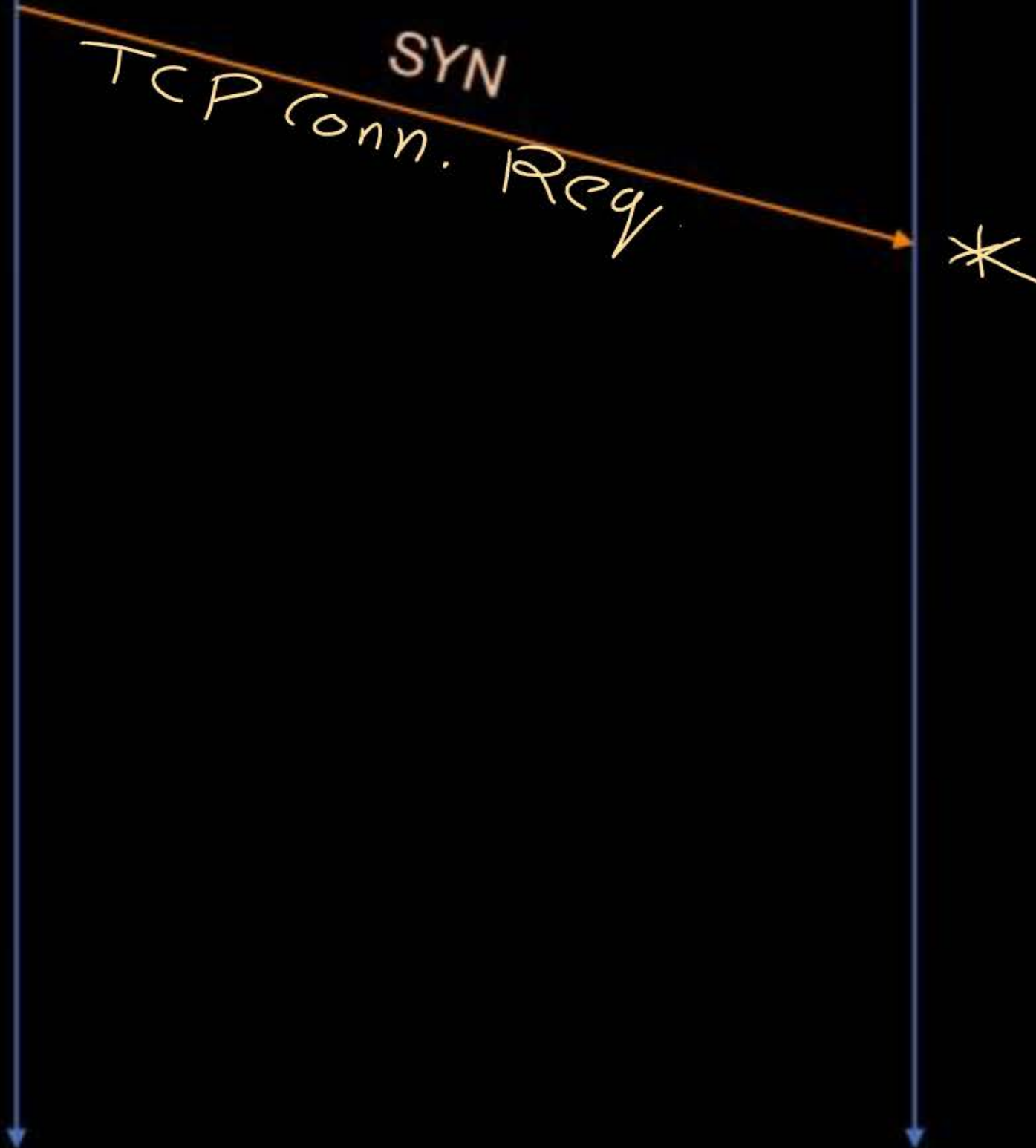


→ 3-way handshake process



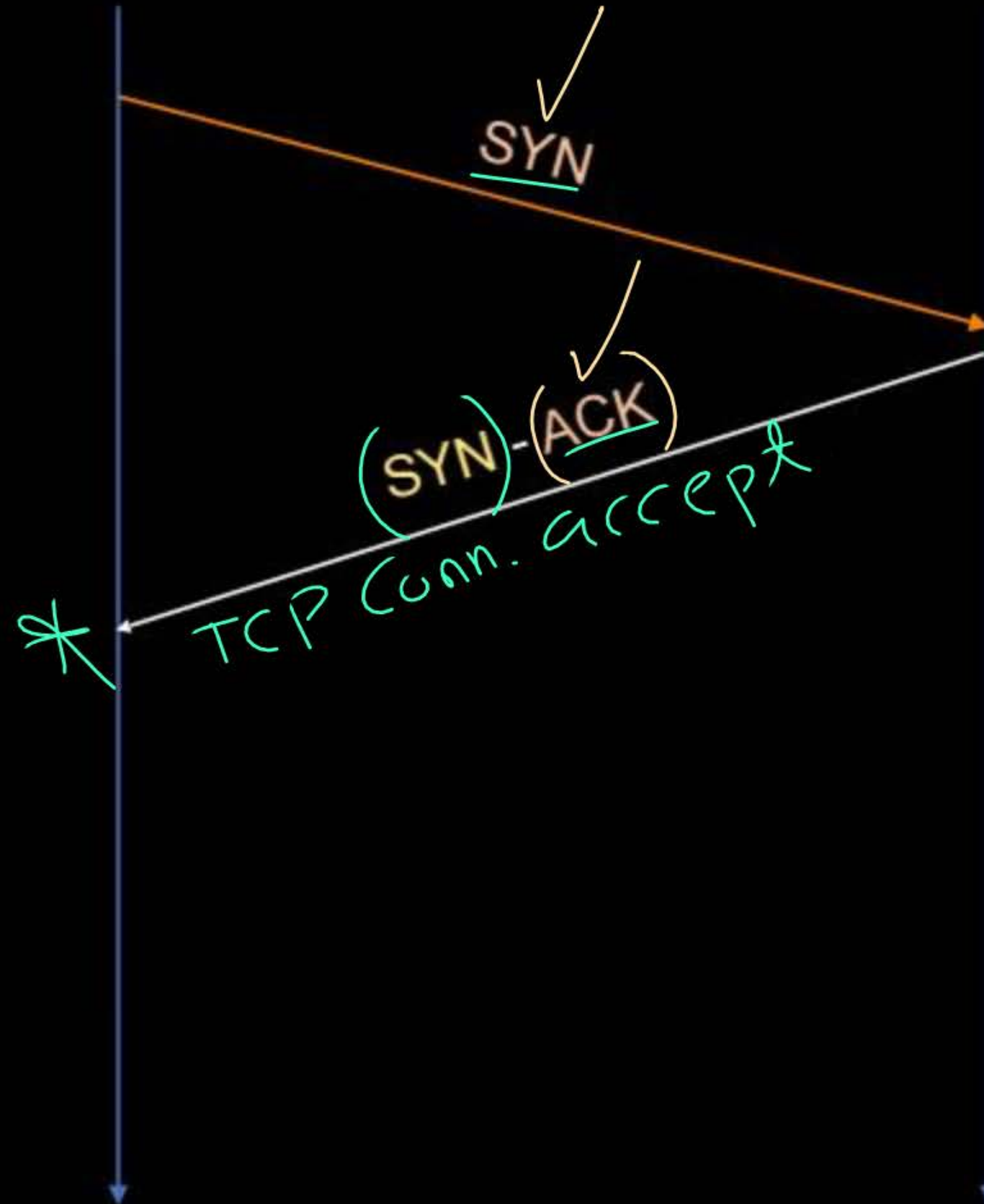
P₁ TCP Client

TCP Server P₂



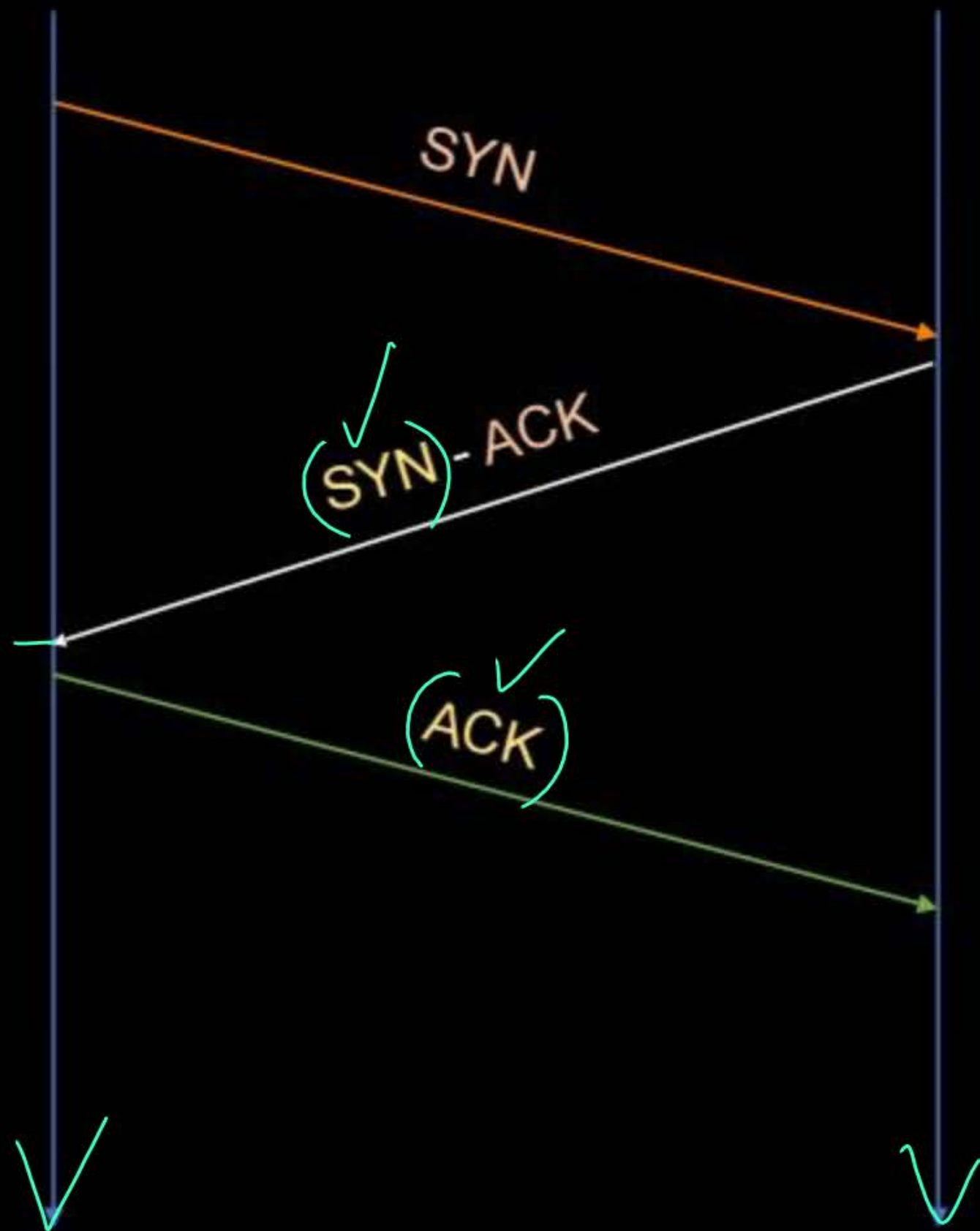
TCP Client

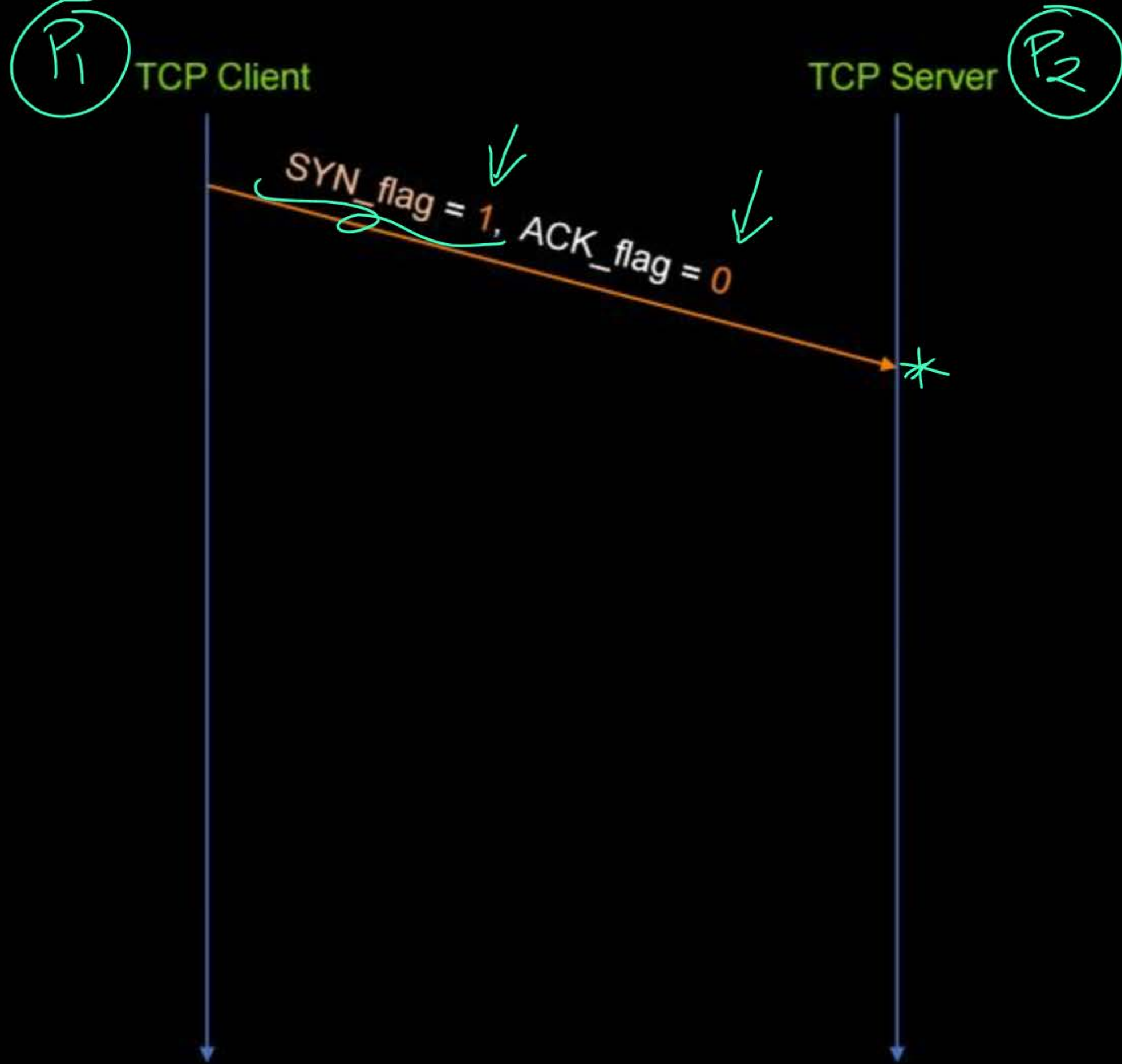
TCP Server



P_1 TCP Client

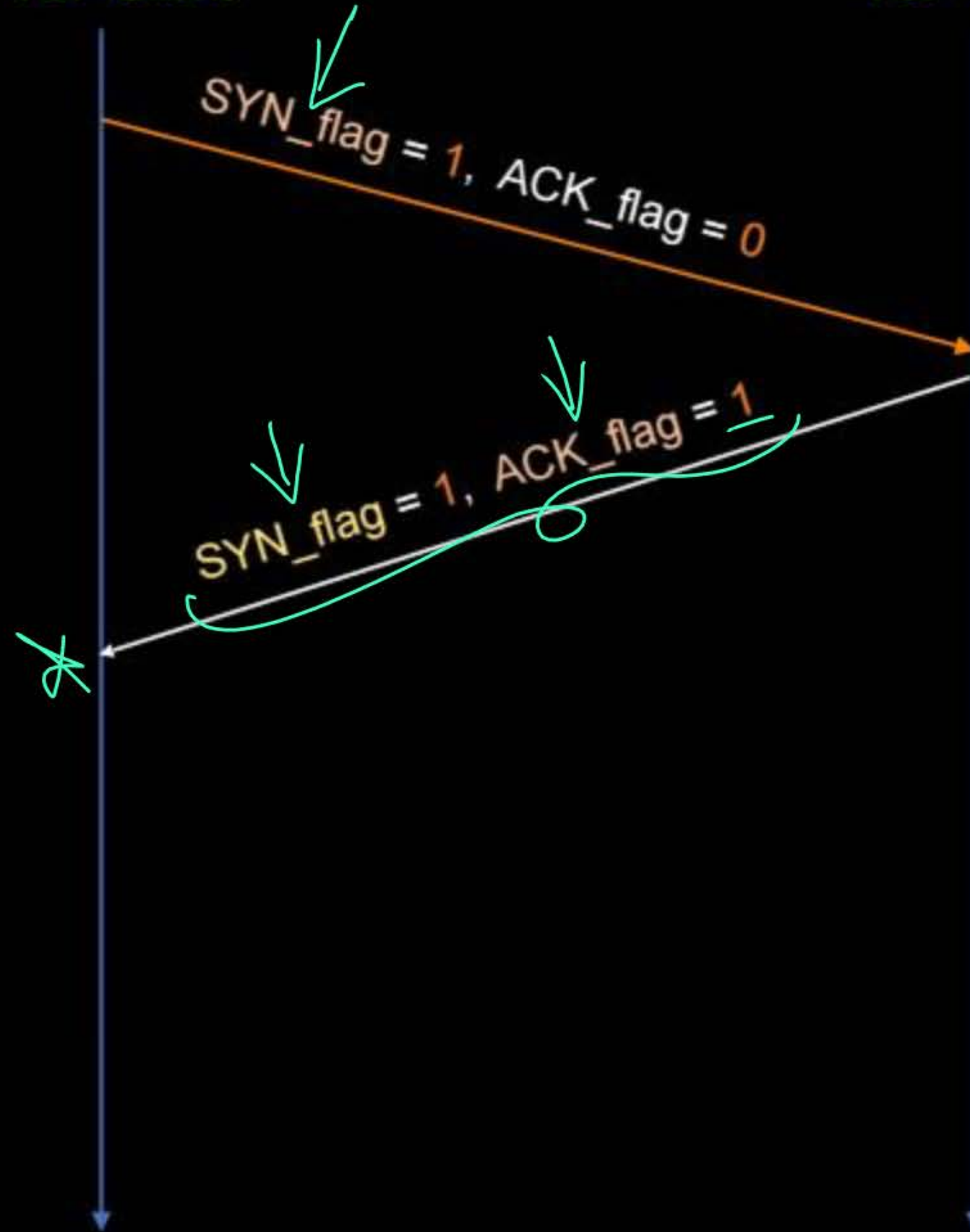
TCP Server P_2





TCP Client

TCP Server

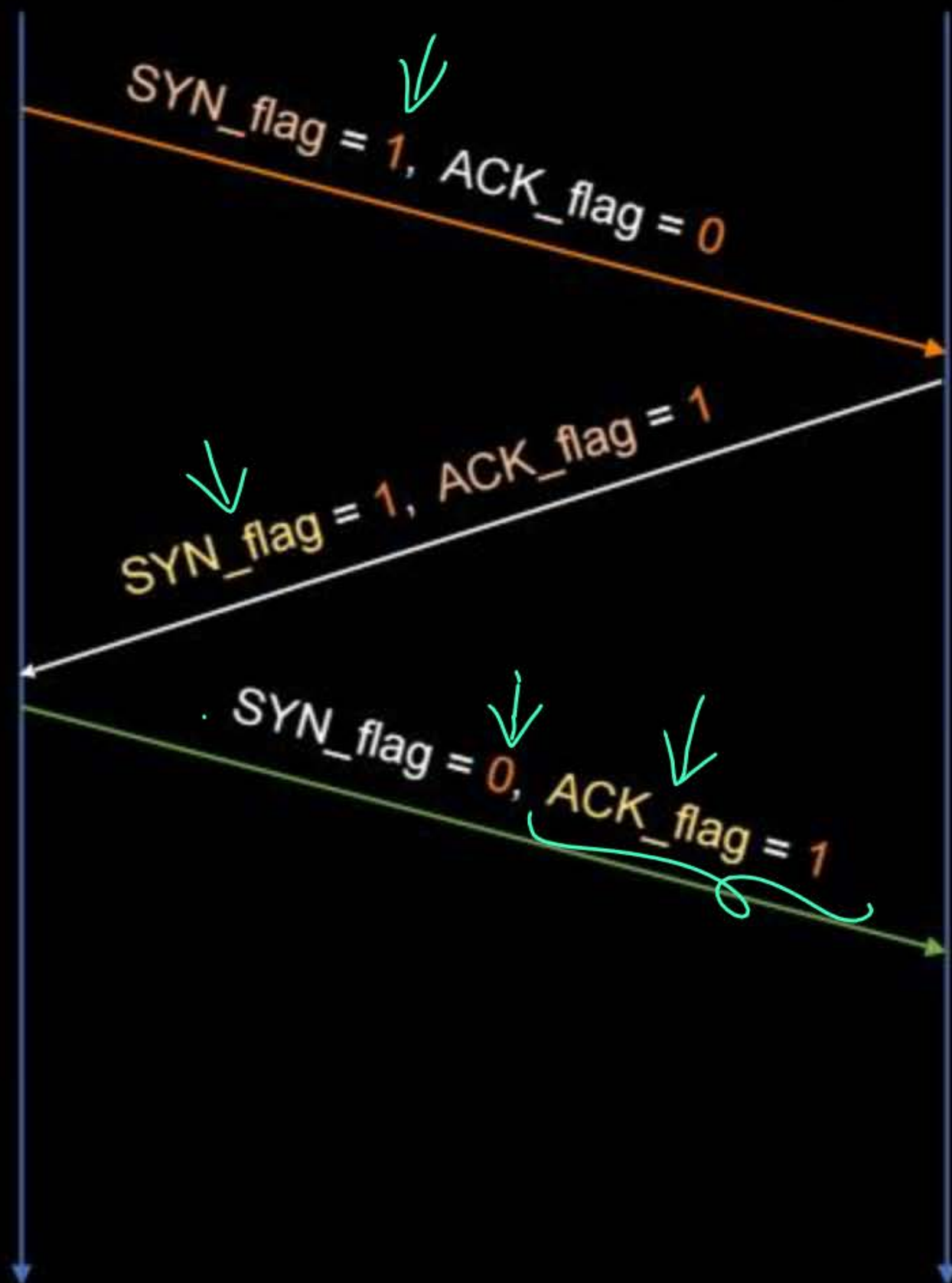


P₁

TCP Client

TCP Server

P₂



[MSQ] [GATE-2025, Set-1] [1-Mark] IIT-R



#Q. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

☒ A P1: SYN = 1 $\rightarrow SYN=1, ACK=0$

☒ B P2: SYN = 1, ACK = 1

☐ C P2: SYN = 0, ACK = 1

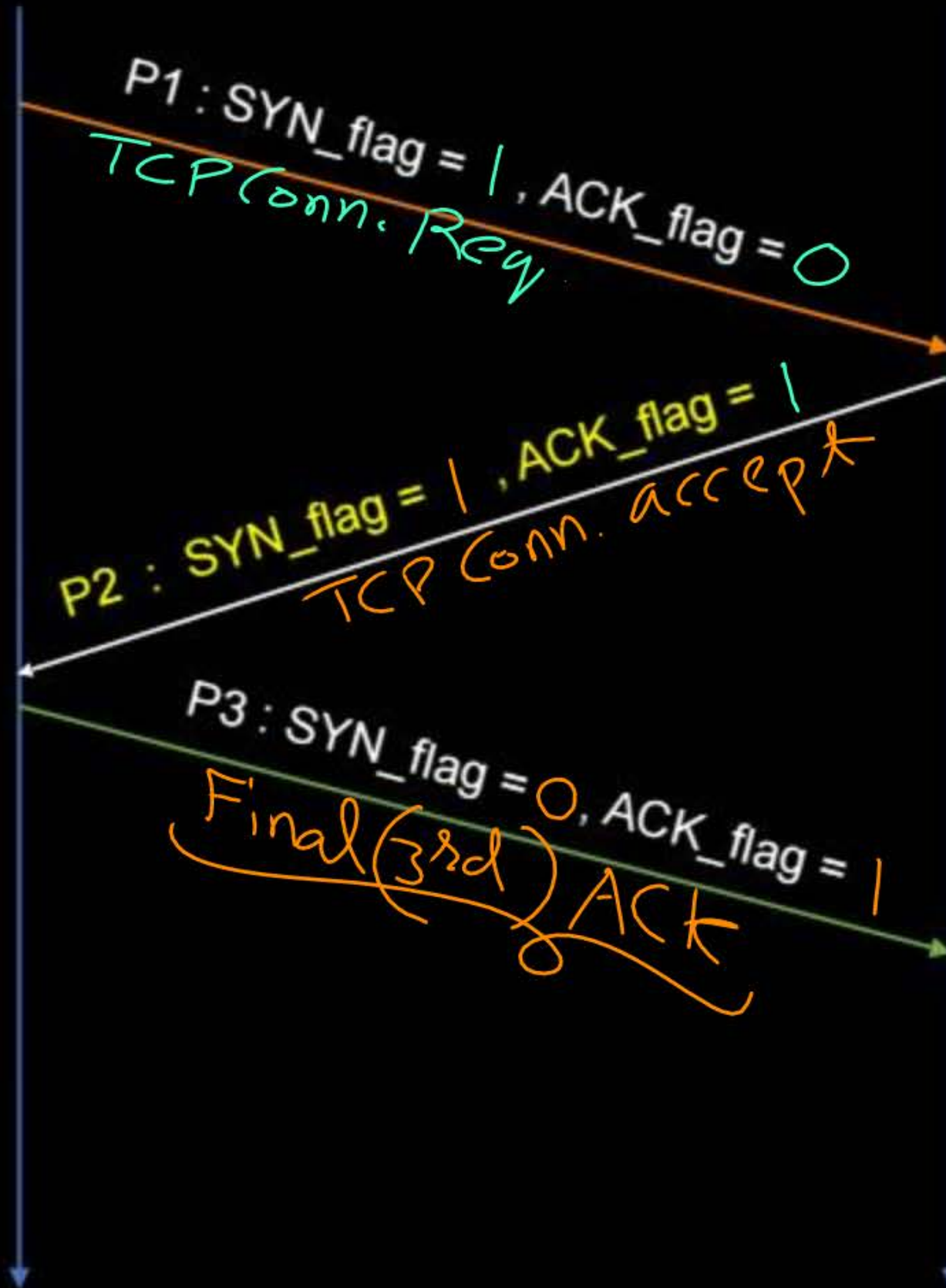
☐ D P3: SYN = 1, ACK = 1

P3

Ans: A & B

TCP Client

TCP Server





Topic

TCP Connection Establishment

① with flag bits ✓

② with seq no. & Ack no. →

⇒ TCP Conn. Close
CS-17, CS-26



THANK - YOU